INSA INSTITUT NATIONAL DES SCIENCES APPLIQUÉES LYON

DPT INFORMATIQUE
COMPUTER SCIENCE AND INFORMATION TECHNOLOGY
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Software Development

Object Oriented Programming - C++ - Advance

IDENTIFICATION

CODE :	IF-3-POO2
ECTS :	2.0

HOURS

Lectures :	9.0 h
Seminars :	6.0 h
Laboratory :	12.0 h
Project :	0.0 h
Teacher-student contact :	27.0 h
Personal work :	25.0 h
Total :	52.0 h

ASSESSMENT METHOD

Lab work evaluation:
- Written report (most of the time).

Final exam:
- Written final exam with documents allowed (duration: one hour and a half).

TEACHING AIDS

Copy of the lecture slides.

TEACHING LANGUAGE

French

CONTACT

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AIMS

The aim of this course is the mastery of the methodological tools and concepts essential to the design, the implementation, the test, and the maintenance of high quality software. The object oriented approach with the C++ programming language is used to reach our goal. This course reinforces the basic knowledge acquired during the teaching module IF-3-POO1 which is a prerequisite for this course.

Skills
Target skills are as follow:
- Applying methodologies for the development of software;
- Designing an object oriented software architecture;
- Designing, implementing and maintaining high quality software.

CONTENT

This course completes the description of the fundamental concepts of the object oriented approach already tackled in the IF-3-POO1 module. It adds: inline, overloading of functions, operators and methods, friendship, namespaces, templates (functions and classes), STL (Standard Template Library), exception handling, standard input/output stream...
At the end of this teaching unit, you should be able:
- To build generic C++ programs (with functions and/or classes);
- To master the inheritance concept (specialization, reuse, polymorphism...) with use of templates;
- To master the use of the STL (Standard Template Library) in particular STL containers, STL algorithms and STL iterators;
- To manipulate the input/output streams with the standard C++ stream library (class hierarchy ios);
- To build and to debug high quality complex object oriented programs using almost all the structure of the C++ programming language.

BIBLIOGRAPHY

[1] Bjarne Stroustrup, The C++ Programming Language (Fourth Edition), Addison-Wesley, 2013, ISBN-13: 978-0321563842
[2] Bjarne Stroustrup, Programming: Principles and Practice Using C++ (Second Edition), Addison-Wesley, 2014, ISBN-13: 978-0321992789
[3] Scott Meyers, Effective Modern C++: 42 Specific Ways to Improve Your Use of C++11 and C++14 (1st Edition), O¿Reilly, 2015, ISBN-13: 978-1491903995
[4] Scott Meyers, Effective C++: 55 Specific Ways to Improve Your Programs and Designs (3rd Edition), Addison-Wesley Professional Computing Series, 2005, ISBN-13: 078-5342334876
[5] Scott Meyers, Effective STL: 50 Specific Ways to Improve Your Use of the Standard Template Library (1st Edition), Addison-Wesley Professional Computing Series, 2008, ISBN-13: 978-0201749625

PRE-REQUISITE

Basic knowledge in algorithmics and in C programming language (IF-3-ALGO).
Object Oriented Programming - C++ - Basis (IF-3-POO1).